THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{3}{4}$	Substitute the standard values
2	**(a)** 20 m	Add the height of the eyes: $2 + 18$; Height of building $= 20$ m
3	**(a)** 30 cm	$r^2 = OP^2 - PT^2 = 34^2 - 16^2 = 900$; $r = 30$ cm
4	**(a)** 56 cm²	Circle $= 154$ cm^2; square $= 98$ cm^2; Difference $= 56$ cm^2
5	**(d)** 3	$\frac{c}{q} + \frac{p}{q}$
6	**(b)** $4\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 8$ gives $h = 8 \div (\frac{2\sqrt{3}}{3})$; Height $= 4\sqrt{3}$ m (distances 12 m and 4 m)
7	**(d)** 92 cm	Perimeter $= PA + AO + OB + BP = 2(16) + 2(30)$; Perimeter $= 92$ cm
8	**(c)** 672 cm²	Circle area $= \frac{22}{7} \times 28^2 = 2464$ cm^2; square $= 3136$ cm^2; Left over $= 672$ cm^2
9	**(a)** $\frac{1}{2}$	$\frac{c}{p} - \frac{q}{p}$
10	**(a)** 24 m	The points are on opposite sides, so the distances add; Distance $= 24$ m
11	**(c)** 360 cm²	Area $= 2 \times \frac{1}{2} \times 40 \times 9$; Area $= 360$ cm^2
12	**(a)** 1034 m²	Area $= \pi(R^2 - r^2) = \frac{22}{7}(27^2 - 20^2)$; Area of path $= 1034$ m^2
13	**(b)** 2	Each ratio is 1
14	**(b)** $3\sqrt{3}$ m	Distance from the foot $= 3\cot 30^\circ = 3\sqrt{3}$ m; Distance from the foot of the tree to the point where its top touches the ground $= 3\sqrt{3}$ m
15	**(a)** 96 cm	Half-chord $= \sqrt{52^2 - 20^2} = 48$ cm; Chord $= 2 \times 48 = 96$ cm

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Divide by $\cos A$: $\frac{\tan A + 1}{\tan A - 1}$ (1) $= 7$ (1).	1+1
17	Horizontal distance $d = 6\cot 60^\circ = 2\sqrt{3}$ m (from the depression of the foot); Part of the tower above the top of the building $= d\tan 30^\circ = 2$ m (1); Tower $= 8$ m (1).	1+1
18	$PA = PB$, so $\triangle PAB$ is isosceles and $\angle PBA = \angle PAB$; $\angle APB = 180^\circ - 30^\circ - 30^\circ$ (1); $\angle APB = 120^\circ$ (1).	1+1
19	In 15 minutes the hand turns through $6^\circ \times 15 = 90^\circ$; Area $= \frac{90}{360} \times \frac{22}{7} \times 28^2$ (1); Area $= 616$ cm^2 (1).	1+1
20	$\frac{1}{2} = \sin 30^\circ$ (1); $2x + 0 = 30$ (1); $x = 15^\circ$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$\frac{1}{2} = \sin 30^\circ$ (1); $3x + 0 = 30$ (1); $x = 10^\circ$ (1).	1+1+1
22	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 4 m (1); $h(\cot 45^\circ - \cot 60^\circ) = 4$ gives $h = 4 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 6 + 2\sqrt{3}$ m (distances $6 + 2\sqrt{3}$ m and $2 + 2\sqrt{3}$ m) (1).	1+1+1
23	Tangents from an external point are equal: $QA = QB$ (1); $9x + 4 = 5x + 20$ gives $x = 4$ (1); $QA = 9(4) + 4 = 40$ cm (1).	1+1+1
24	Circumference $= 2\pi r = 2 \times \frac{22}{7} \times 14 = 88$ m (1); Cost $= 88 \times 12$ (1); Cost $=$ ₹ 1056 (1).	1+1+1
	OR: Hypotenuse $= \sqrt{15^2 + 8^2} = 17$ (1); $\sin\theta = \frac{15}{17}$, $\cos\theta = \frac{8}{17}$ (1); $\sec\theta = \frac{17}{8}$ (1).	1+1+1
25	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 6\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 6\sqrt{3}$, so $d = 9$ (1); $H = d\tan 60^\circ = 9\sqrt{3}$; tower $9\sqrt{3}$ m, distance 9 m (1).	1+1+1
	OR: For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $18 + 12 = AD + 15$ (1); $AD = 15$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	LHS $= \left(\dfrac{1-\sin A}{\cos A}\right)^2$ (2); $= \dfrac{(1-\sin A)^2}{1-\sin^2 A}$ (1); $= \dfrac{1-\sin A}{1+\sin A}$ (2).	5
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 60 (2); $h\left(\frac{1}{\sqrt{3}}+\sqrt{3}\right) = 60$ gives $h = 15\sqrt{3}$ (2); Height $15\sqrt{3}$ m; distances 15 m and 45 m (1).	2+2+1
27	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{52^2 - 20^2} = 48$ cm; perimeter $= 2(48+20)$ (2); Perimeter 136 cm; area $= 2 \times \frac{1}{2} \times 20 \times 48 = 960$ cm^2 (1).	2+2+1
	OR: Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{45}{360} \times \frac{22}{7} \times (14^2 - 7^2)$ (2); Area $= 57.75$ cm^2 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 77 cm²	$\frac{60}{360} \times \frac{22}{7} \times (14^2 - 7^2)$; Area $= 77$ cm^2
2	**(b)** $\frac{7}{3}$	Divide by $\cos\theta$
3	**(b)** $3 + 2\sqrt{3}$ m	Distance from the foot $= 3\cot 60^\circ = \sqrt{3}$ m; Original height of the tree $= 3 + 2\sqrt{3}$ m
4	**(d)** 5 cm	$AF = AB - r = 11 - r$, $CD = BC - r = 60 - r$ and $AF + CD = AC$ (tangent lengths); $(11 - r) + (60 - r) = 61$ gives $r = 5$ cm
5	**(c)** 90°	$\theta = \frac{22\times 360}{2\times\frac{22}{7}\times 14}$; $\theta = 90^\circ$
6	**(a)** 70°	$\cos(90^\circ - \theta) = \sin\theta$
7	**(b)** $5\sqrt{3}$ m	$h = 10\sin 60^\circ = 10 \times \frac{\sqrt{3}}{2}$; Height $= 5\sqrt{3}$ m
8	**(c)** 43 cm	$8x + 3 = 5x + 18$ gives $x = 5$; $QA = 8(5) + 3 = 43$ cm
9	**(c)** 686 cm²	Sector $= 1886.5$ cm^2, triangle $= \frac{1}{2} \times 49 \times 49 = 1200.5$ cm^2; Segment $= 1886.5 - 1200.5 = 686$ cm^2
10	**(c)** $\frac{7}{6}$	Substitute the standard values
11	**(c)** $7 + 7\sqrt{3}$ m	$p + 14 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 14$; $p = \frac{14}{\sqrt{3}-1} = 7 + 7\sqrt{3}$ m
12	**(d)** 110°	$\angle APB = 180^\circ - 35^\circ - 35^\circ$; $\angle APB = 110^\circ$
13	**(d)** 1089 cm²	Side $= 132 \div 4 = 33$ cm; Area $= 33^2 = 1089$ cm^2
14	**(c)** 5	$\frac{c}{q} + \frac{p}{q}$
15	**(d)** one-third of the distance of P from the foot	$\cot 60^\circ : \cot 30^\circ = 1 : 3$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Height of the top above the level of his eyes $= 24\tan 60^\circ = 24\sqrt{3}$ m; Add the height of the eyes: $2 + 24\sqrt{3}$ (1); Height of building $= 2 + 24\sqrt{3}$ m (1).	1+1
17	$PA = PB$, so $\angle PAB = \frac{180^\circ - 36^\circ}{2} = 72^\circ$; $\angle OAP = 90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB = 90^\circ - 72^\circ = 18^\circ$ (1).	1+1
18	The horse can graze a sector of angle 180° and radius 14 m; Area $= \frac{180}{360} \times \frac{22}{7} \times 14^2$ (1); Area $= 308$ m^2 (1).	1+1
19	Use $\sin^2 45^\circ = \frac{1}{2}$, $\cos^2 45^\circ = \frac{1}{2}$ (2); value $= 1$ (1).	2+1
20	Distances from the foot: $h\cot 45^\circ = 9$ m (farther) and $h\cot 60^\circ = 3\sqrt{3}$ m (nearer); Both objects are on the same side, so the distances are subtracted (1); Distance $= 9 - 3\sqrt{3}$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Let the new radius be R; $2\pi R = 2\pi r_1 + 2\pi r_2$ (1); $R = r_1 + r_2 = 14 + 7$ (1); $R = 21$ cm (1).	1+1+1
22	$\sqrt{\cos^2 A} = \cos A$ (1); $\cos A \cdot \sec A$ (1) $= 1$ (1).	1+1+1
23	Distance of the point from the building: $d = 6\cot 45^\circ = 6$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 6\sqrt{3}$ m (1); Length $= 6\sqrt{3} - 6 = 6\sqrt{3} - 6$ m (1).	1+1+1
24	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 30 \times 16$ (1); Area $= 480$ cm^2 (1).	1+1+1
	OR: Let the new radius be R; $\pi R^2 = \pi r_1^2 + \pi r_2^2$ (1); $R^2 = 3^2 + 4^2 = 25$ (1); $R = 5$ cm (1).	1+1+1
25	$\sqrt{3} = \tan 60^\circ$ (1); $2x + 0 = 60$ (1); $x = 30^\circ$ (1).	1+1+1
	OR: Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 8 m (1); $h(\cot 45^\circ - \cot 60^\circ) = 8$ gives $h = 8 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 12 + 4\sqrt{3}$ m (distances $12 + 4\sqrt{3}$ m and $4 + 4\sqrt{3}$ m) (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Divide numerator and denominator by $\cos A$ (1): $\dfrac{3\tan A+2}{4\tan A+1}$ (1); $=\dfrac{3(\frac{15}{8})+2}{4(\frac{15}{8})+1}=\frac{61}{68}$ (2); verified with $\sin A=\frac{15}{17}$, $\cos A=\frac{8}{17}$ (1).	1+1+2+1
	OR: Let height $=h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 45^\circ$, which differ by 2 m (2); $h(\cot 30^\circ-\cot 45^\circ)=2$ gives $h=2\div(\sqrt{3}-1)$ (2); Height $=1+\sqrt{3}$ m (distances $3+\sqrt{3}$ m and $1+\sqrt{3}$ m) (1).	2+2+1
27	$OA\perp PA$ and $OB\perp PB$, so $\angle OAP=\angle OBP=90^\circ$ (2); In quadrilateral $OAPB$ the angle sum is 360° (1); $\angle APB+\angle AOB=360^\circ-90^\circ-90^\circ=180^\circ$ (2).	2+1+2
	OR: The 8 ribs divide the circle into 8 equal sectors (2); Area of the circle $=\frac{22}{7}\times 42^2=5544$ cm^2 (2); Area between ribs $=5544\div 8=693$ cm^2 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 13 cm	$OP^2 = 144 + 25 = 169$
2	**(b)** 42 cm²	Quadrant $= 154$ cm^2; square $= 196$ cm^2; Remaining $= 42$ cm^2
3	**(d)** $\frac{2}{3}$	$\frac{c}{p} - \frac{q}{p}$
4	**(b)** $5 + 8\sqrt{3}$ m	Add the height of the eyes: $5 + 8\sqrt{3}$; Height of building $= 5 + 8\sqrt{3}$ m
5	**(b)** 28 cm	$r^2 = OP^2 - PT^2 = 35^2 - 21^2 = 784$; $r = 28$ cm
6	**(c)** 224 cm²	Circle $= 616$ cm^2; square $= 392$ cm^2; Difference $= 224$ cm^2
7	**(c)** 2	Each ratio is 1
8	**(d)** $6\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 12$ gives $h = 12 \div (\frac{2\sqrt{3}}{3})$; Height $= 6\sqrt{3}$ m (distances 18 m and 6 m)
9	**(c)** 98 cm	Perimeter $= PA + AO + OB + BP = 2(21) + 2(28)$; Perimeter $= 98$ cm
10	**(d)** 1050 cm²	Circle area $= \frac{22}{7} \times 35^2 = 3850$ cm^2; square $= 4900$ cm^2; Left over $= 1050$ cm^2
11	**(d)** $\frac{17}{5}$	Divide by $\cos\theta$
12	**(d)** 48 m	The points are on opposite sides, so the distances add; Distance $= 48$ m
13	**(a)** 972 cm²	Area $= 2 \times \frac{1}{2} \times 36 \times 27$; Area $= 972$ cm^2
14	**(c)** 814 m²	Area $= \pi(R^2 - r^2) = \frac{22}{7}(22^2 - 15^2)$; Area of path $= 814$ m^2
15	**(b)** 55°	$\cos(90^\circ - \theta) = \sin\theta$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 52^2 - 48^2 = 400$ (1); $PT = 20$ cm (1).	1+1
17	Circumference $= 2 \times \frac{22}{7} \times 35 = 220$ cm; Distance 55 m $= 5500$ cm; revolutions $= \frac{5500}{220}$ (1); Revolutions $= 25$ (1).	1+1
18	Divide by $\cos A$: $\frac{\tan A + 1}{\tan A - 1}$ (1) $= \frac{17}{7}$ (1).	1+1
19	Horizontal distance $d = 12\cot 60^\circ = 4\sqrt{3}$ m (from the depression of the foot); Part of the tower above the top of the building $= d\tan 30^\circ = 4$ m (1); Tower $= 16$ m (1).	1+1
20	$PA = PB$, so $\triangle PAB$ is isosceles and $\angle PBA = \angle PAB$; $\angle APB = 180^\circ - 55^\circ - 55^\circ$ (1); $\angle APB = 70^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Tangents from an external point are equal: $QA = QB$ (1); $7x + 2 = 4x + 14$ gives $x = 4$ (1); $QA = 7(4) + 2 = 30$ cm (1).	1+1+1
22	Circumference $= 2\pi r = 2 \times \frac{22}{7} \times 21 = 132$ m (1); Cost $= 132 \times 9$ (1); Cost $=$ ₹ 1188 (1).	1+1+1
23	Hypotenuse $= \sqrt{24^2 + 7^2} = 25$ (1); $\sin\theta = \frac{24}{25}$, $\cos\theta = \frac{7}{25}$ (1); $\sec\theta = \frac{25}{7}$ (1).	1+1+1
24	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 8\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 8\sqrt{3}$, so $d = 12$ (1); $H = d\tan 60^\circ = 12\sqrt{3}$; tower $12\sqrt{3}$ m, distance 12 m (1).	1+1+1
	OR: For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $21 + 14 = AD + 17$ (1); $AD = 18$ cm (1).	1+1+1
25	Let the new radius be R; $2\pi R = 2\pi r_1 + 2\pi r_2$ (1); $R = r_1 + r_2 = 18 + 11$ (1); $R = 29$ cm (1).	1+1+1
	OR: $\frac{\sin A}{\csc A} = \sin^2 A$, $\frac{\cos A}{\sec A} = \cos^2 A$ (2); sum $= 1$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	LHS $= \dfrac{\cos^2 A + (1+\sin A)^2}{\cos A(1+\sin A)}$ (2); $= \dfrac{2+2\sin A}{\cos A(1+\sin A)}$ (2); $= \dfrac{2}{\cos A} = 2\sec A$ (1).	5
	OR: Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 100 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 100$ gives $h = 25\sqrt{3}$ (2); Height $25\sqrt{3}$ m; distances 25 m and 75 m (1).	2+2+1
27	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{53^2 - 28^2} = 45$ cm; perimeter $= 2(45+28)$ (2); Perimeter 146 cm; area $= 2 \times \frac{1}{2} \times 28 \times 45 = 1260$ cm^2 (1).	2+2+1
	OR: Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{72}{360} \times \frac{22}{7} \times (14^2 - 7^2)$ (2); Area $= 92.4$ cm^2 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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